

Schrödinger Bridge for Non-linear ROMs

Optimal Transport with Hessian-Based Metric Densities

Presentation of Progress

June 26, 2026

Feature Extraction: Limitations of Standard Sensors

- Standard sensors (like Ducros) rely heavily on the velocity divergence ($\nabla \cdot \vec{v} < 0$) to identify compressive shock waves.
- **The Issue:** Expansion fans (ondes de détente) are C^1 discontinuities characterized by decompression ($\nabla \cdot \vec{v} > 0$). Kinematic sensors explicitly delete them.
- Consequently, the Schrödinger Bridge transports only the shocks, leaving the expansion fans behind as un-transported numerical artifacts.

Objective

Construct a probability marginal $\mu(x)$ that captures both strong C^0 shock waves and weak C^1 expansion fans simultaneously, directly on the unstructured mesh.

Alauzet & Loseille (2011) Framework

We construct a probability density by computing the node density of the L^p -optimal anisotropic Riemannian metric tensor.

The Absolute Hessian of the Mach field $M(x)$ is computed via two-pass FVM Least-Squares:

$$|H_M| = R \begin{bmatrix} |\lambda_1| & 0 \\ 0 & |\lambda_2| \end{bmatrix} R^T$$

The probability marginal $\mu(x)$ is defined by the fractional-power metric determinant:

$$\mu(x) \propto (\det |H_M|)^{\frac{p}{2p+d}} = (|\lambda_1 \lambda_2|)^{\frac{p}{2p+2}}$$

Hessian Modes: det vs eig

We implemented two variations of the metric feature density:

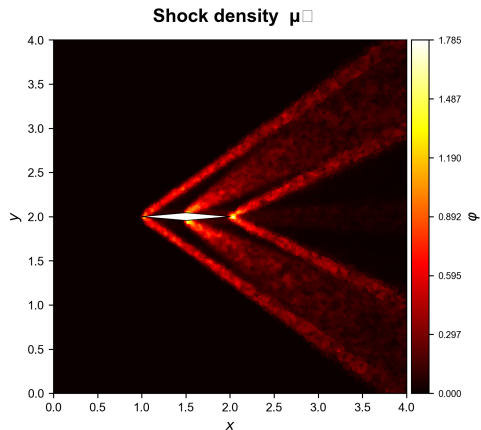
- **det mode (Strict Metric Density):**

$$\mu(x) \propto \left| \frac{\partial^2 M}{\partial x^2} \frac{\partial^2 M}{\partial y^2} - \left(\frac{\partial^2 M}{\partial x \partial y} \right)^2 \right|^{\frac{p}{2p+2}}$$

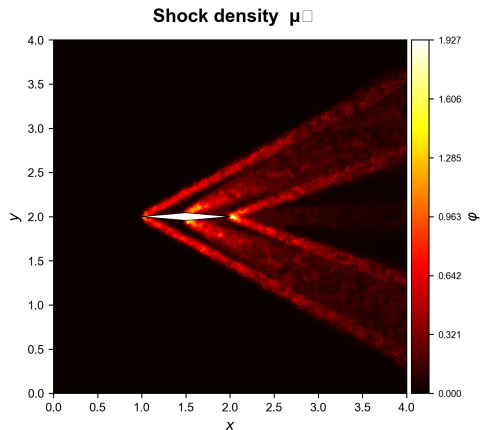
- **eig mode (Isotropic Feature Strength):**

$$\mu(x) \propto |\lambda_{\max}|^{\frac{p}{2p+2}}$$

Results: Hessian Densities (Diamond Body)



μ_0 — Mach 2.00 (*det mode*)

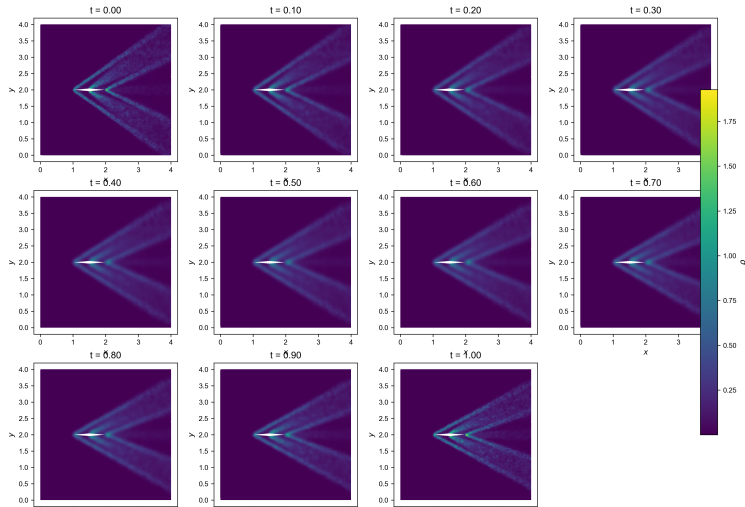


μ_1 — Mach 2.50 (*det mode*)

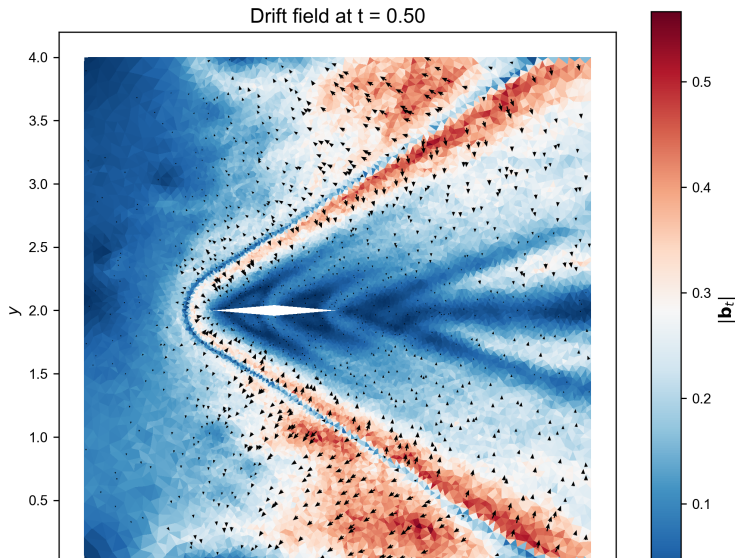
Both the leading/trailing shocks and the expansion corners are successfully highlighted.

Schrödinger Bridge Density Transport (Diamond Body)

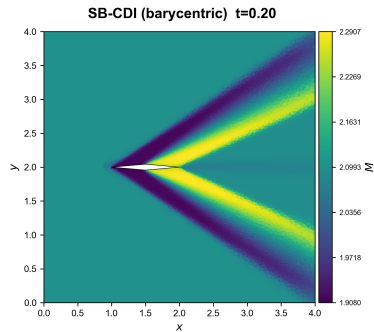
2D density transport



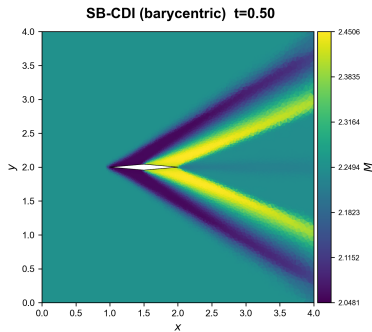
Schrödinger Bridge Drift Field (Diamond Body)



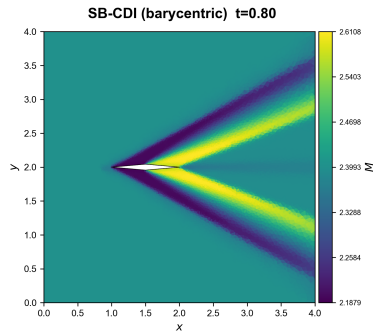
Mach Interpolation Trajectory (Diamond Body, det mode)



$t = 0.20$

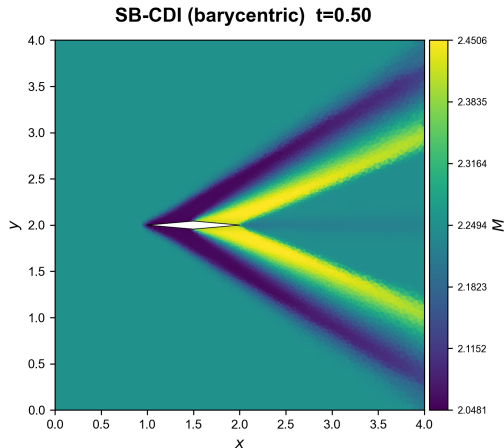


$t = 0.50$

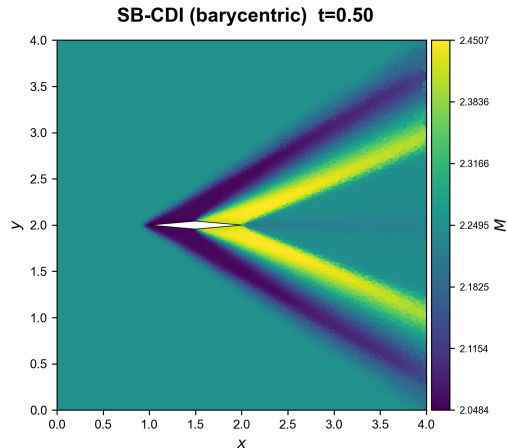


$t = 0.80$

Mach Interpolation at $t = 0.5$ (Diamond Body)

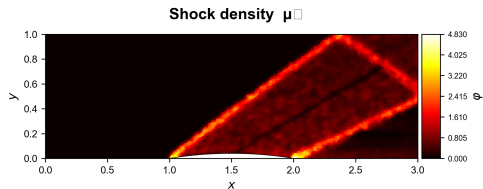


Barycentric CDI (det mode)

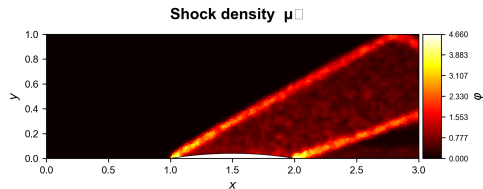


Barycentric CDI (eig mode)

Results: Hessian Densities (Transonic Bump)



μ_0 — Mach 1.20 (*det mode*)

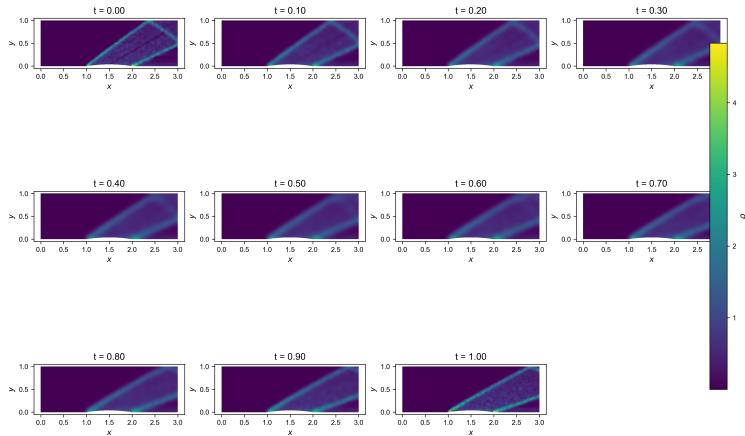


μ_1 — Mach 1.30 (*det mode*)

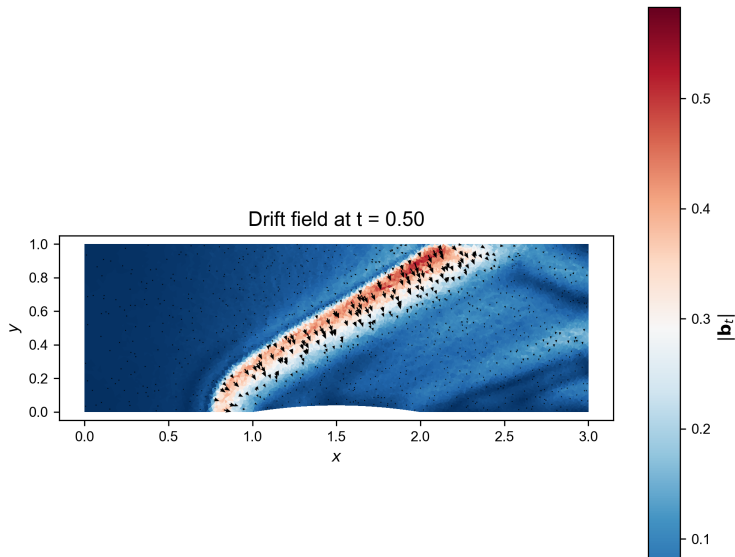
The Hessian isolates both the λ -shock and the continuous expansion zone over the crest.

Schrödinger Bridge Density Transport (Bump)

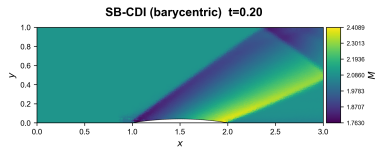
2D density transport



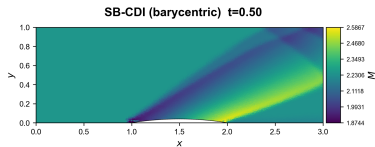
Schrödinger Bridge Drift Field (Bump)



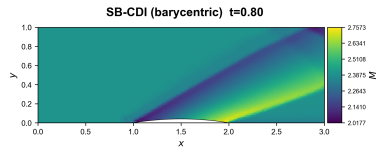
Mach Interpolation Trajectory (Transonic Bump, det mode)



$t = 0.20$

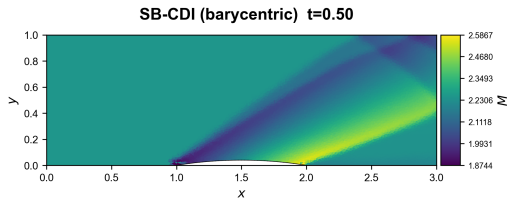


$t = 0.50$

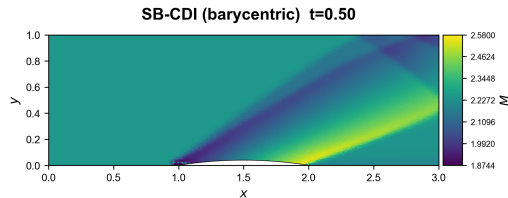


$t = 0.80$

Mach Interpolation at $t = 0.5$ (Transonic Bump)



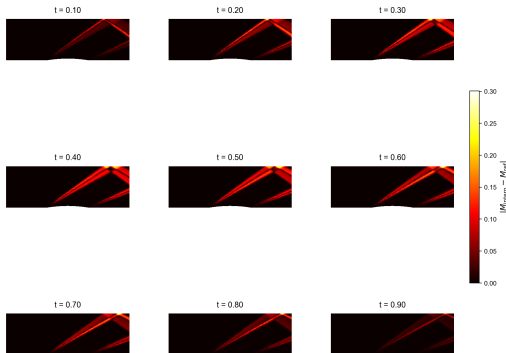
Barycentric CDI (det mode)



Barycentric CDI (eig mode)

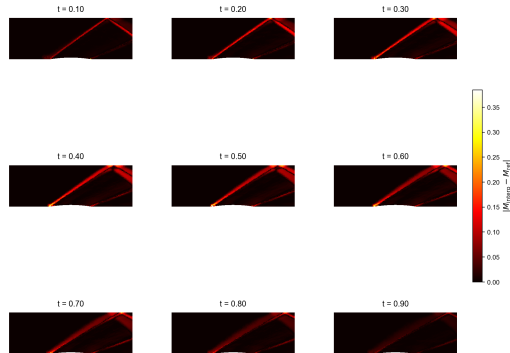
Error vs. High-Resolution Reference (Bump)

$|\text{Linear} - \text{ref}|$



Absolute Error: Linear Interpolation

$|\text{BaryCDI} - \text{ref}|$



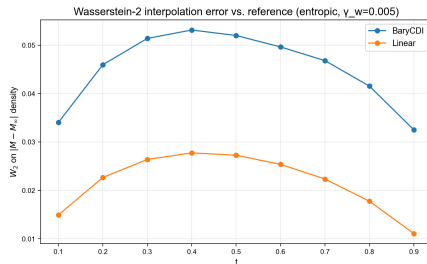
Absolute Error: Barycentric CDI (det)

Quantitative Error: Wasserstein-2 Metrics (4 Cases)

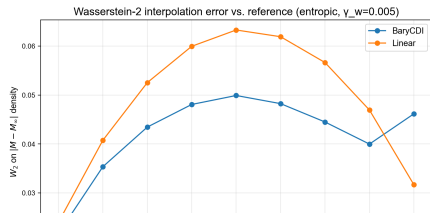
Diamond Body



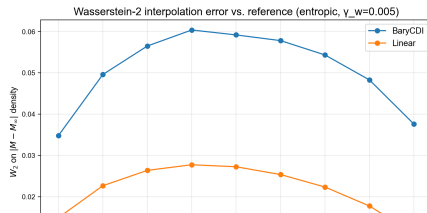
Transonic Bump



det mode



det mode



Quantitative Error Metrics (Bump, det mode)

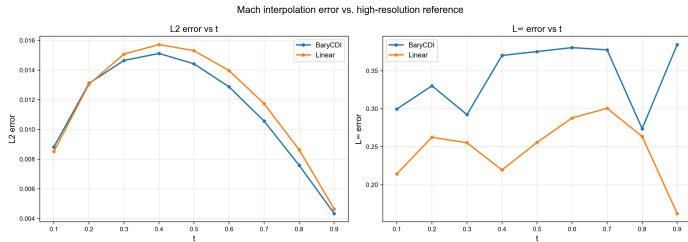


Figure: L_2 and L_∞ Errors vs High-Res Ref